

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2021/2022

MAY EXAMINATION

UCCD2323 FRONT-END WEB DEVELOPMENT /
UCCD2223 WEB SITE DESIGN AND DEVELOPMENT

THURSDAY, 19 MAY 2022

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING

Instruction to Candidates :

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in section B, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are not allowed to use calculator.

Answer questions only in the answer booklet provided.

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Section A (Answer ALL Questions)

- Q1. (a) List **FOUR (4)** different types of web hosting. (4 marks)
- (b) With the aid of diagrams, compare any **TWO (2)** different types of web hosting listed in Q1 (a). (10 marks)
- (c) Write the markup to create the table with the design as shown in **Figure 1** below.

9am - 10am	10am - 11am	Schedule
Lecture		

Figure 1: Teaching Timetable

(11 marks)
[Total : 25 marks]

- Q2. (a) Write the HyperText Markup Language (HTML) and the embedded Cascading Style Sheet (CSS) also known as internal or document style sheet to create the content shown in **Figure 2**.

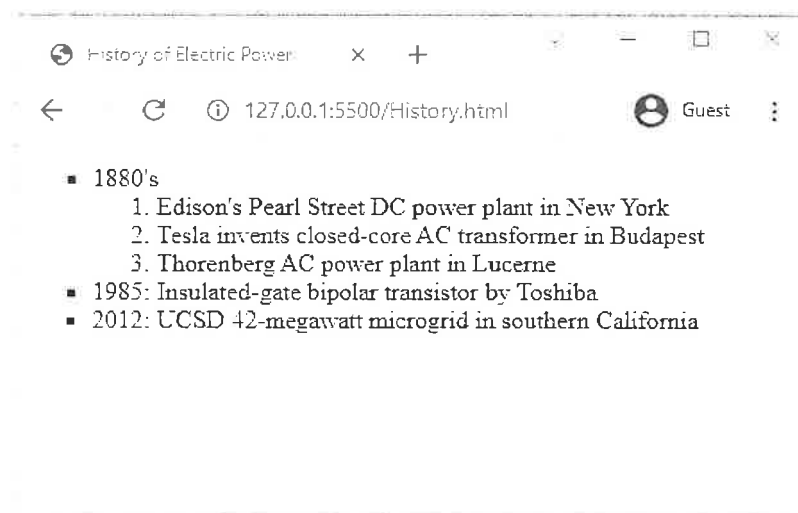


Figure 2: History of Electric Power

All odd numbered items in 1880's for example Item 1 and Item 3 have the background colour set to **yellow** while all even numbered items in 1880's for example Item 2 has the background color set to **lightcyan**. (13 marks)

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Q2. (Continued)

- (b) A web page has the HTML markup as shown below:

```
<body>
  <h1>External Link</h1>
  <p>
    <a href="http://www.utar.edu.my">UTAR</a>
  </p>
</body>
```

Write the embedded CSS so that the hyperlink UTAR is not underlined but when the mouse is moved over the hyperlink, it's text color will turn to red and the size of the font will appear larger. (6 marks)

- (c) The traditional layout is as shown in **Figure 3**. Draw the layout that uses the HTML5 layout elements to represent the traditional layout.

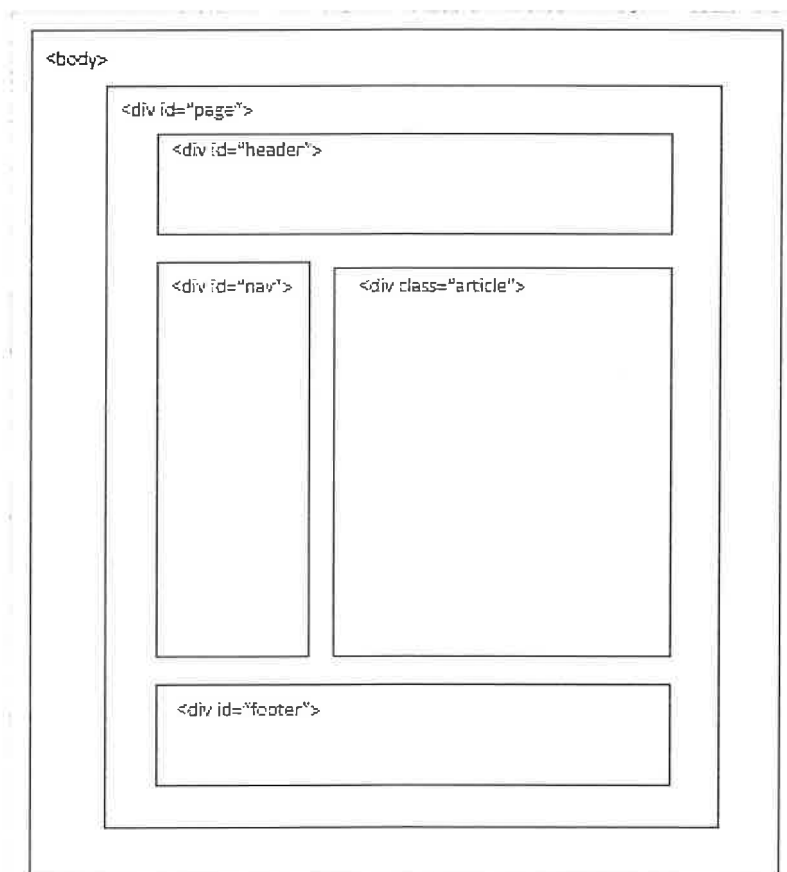


Figure 3: Traditional layout

(6 marks)
 [Total : 25 marks]

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Q3 (a) Provide **THREE (3)** naming rules for a variable in JavaScript. (6 marks)

(b) A JSON object as shown in **Figure 4** contains the social media of SammyShark. Write the JavaScript code to display the social media of SammyShark in a table. The output is as shown in **Figure 5**. When the link is clicked, an alert window will appear to display the value for “about” as shown in **Figure 6**.

```
var user_profile = {
  "username": "SammyShark",
  "social_media": [
    {
      "description": "twitter",
      "link": "https://twitter.com/digitalocean",
      "about": "DigitalOcean is the easiest platform to deploy, manage & scale applications of any size."
    },
    {
      "description": "facebook",
      "link": "https://www.facebook.com/DigitalOceanCloudHosting",
      "about": "DigitalOcean simplifies cloud computing so developers and businesses can spend more time building software that changes the world."
    },
    {
      "description": "github",
      "link": "https://github.com/digitalocean",
      "about": "There are 20 members in DigitalOcean's GitHub."
    }
  ]
}
```

Figure 4: A JSON object

Description	link
twitter	https://twitter.com/digitalocean
facebook	https://www.facebook.com/DigitalOceanCloudHosting
github	https://github.com/digitalocean

Figure 5: Output of SammyShark's social media info

Figure 6: The value for “About” is displayed in an alert window when the link <https://github.com/digitalocean> is clicked

(8 marks)

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Q3. (Continued)

- (c) The markup in **Figure 7** produces the web page as shown in **Figure 8**.

```
<div>
  <p>
    Input Year: <input type="text" id="year" /> <br>
    <input type="button" id="button" onClick="isLeapYear()"
    value="Check Leap Year">
  </p>
  <p id="result"></p>
</div>
```

Figure 7: Markup for a web page that checks for leap year

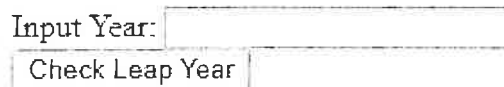


Figure 8: Output of the web page that checks for leap year

When the “Check Leap Year” button is clicked and if the year entered into the text box is a leap year, the string “Is a leap year!” will be displayed otherwise if the year entered into the text box is not a leap year then the string “Not a leap year!” will be displayed. Refer to **Figure 9** and **Figure 10** for the sample outputs.

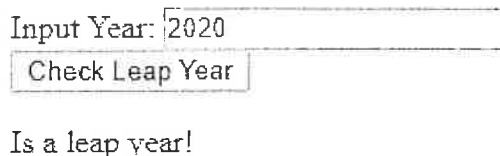


Figure 9: Output for leap year, eg. 1940, 1996, 2000, 2020, ...

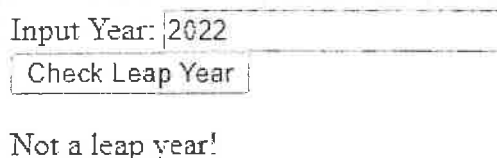


Figure 10: Output for a year that is not a leap year

Write the JavaScript code to implement the function “isLeapYear” that satisfies the criteria mentioned above.

[Note: Leap years are either divisible by 4 but not by 100 or are divisible by 400.]

(11 marks)

[Total : 25 marks]

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Section B (Answer Only ONE (1) Question)

- Q4. (a) Demonstrate the use of **THREE (3)** primitive data types that are supported by JavaScript. Support your answer for the respective category with JavaScript statement(s). (6 marks)
- (b) Briefly explain and demonstrate the use of the <noscript> tag. (7 marks)
- (c) Discuss and show **ONE (1)** programming solution for storing the JSON object shown in **Figure 11** without the use of a database so that the stored info can be retrieved from different browser sessions. [Note: Include code snippets for storing the JSON object and retrieving the stored info.]

```
var user_profile = {
  "username": "SammyShark",
  "social_media": [
    {
      "description": "twitter",
      "link": "https://twitter.com/digitalocean",
      "about": "DigitalOcean is the easiest platform to deploy, manage & scale applications of any size."
    },
    {
      "description": "facebook",
      "link": "https://www.facebook.com/DigitalOceanCloudHosting",
      "about": "DigitalOcean simplifies cloud computing so developers and businesses can spend more time building software that changes the world."
    },
    {
      "description": "github",
      "link": "https://github.com/digitalocean",
      "about": "There are 20 members in DigitalOcean's GitHub."
    }
  ]
};
```

Figure 11: A JSON Object

(12 marks)
[Total : 25 marks]

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- Q5. (a) Write HTML markup to embed a video in a web page. The video source files is "horse.mp4". Users can choose to display the caption in Chinese. The caption file is "horse-zh.vtt". (9 marks)
- (b) With the aid of code snippets, contrast between the global variable scope and local variable scope in JavaScript. (10 marks)
- (c) Briefly describe **THREE (3)** problems with the use of Cookies. (6 marks)
- [Total : 25 marks]
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